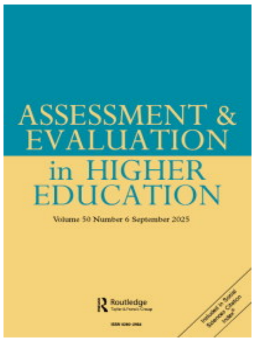

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A systematic-narrative review of performance-based assessments of critical thinking in higher education

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ABSTRACT
Critical thinking encompasses a complex interplay of cognitive skills, knowledge, and dispositions. In this systematic-narrative review, we analyze performance-based assessments of critical thinking in higher education. The aim is to explore how prior peer-reviewed empirical studies conceptualize and assess critical thinking, the methods and frameworks employed, and the specific aspects measured. A total of 28 articles from 2013 to 2023 were included in the analysis. The findings indicate that the majority of studies utilized open-ended performance tasks to measure critical thinking, while selected-response tasks were less common. There was no consistent theoretical framework referenced across the studies. Most studies defined critical thinking as a skill or a set of skills, with some also emphasizing the importance of disposition. The skill-based perspective was also emphasized in the measured aspects, with evaluation and inference as the most frequently assessed skills. None of the reviewed studies explicitly expressed dispositions in their targets of assessment. The results revealed misalignment between the definition of critical thinking and performance-based assessments of critical thinking. Future directions for these assessments are discussed.

KEYWORDS
Performance-based assessment; critical thinking; systematic-narrative review

Introduction

Critical thinking is considered one of the key objectives of higher education. Although the definition of critical thinking has been widely debated in the academic community, clear consensus remains elusive (Butler 2024; Liu, Frankel, and Roohr 2014). Critical thinking can be conceptualized as a complex combination of cognitive skills, knowledge, and affective dispositions (Halpern 2014). These cognitive skills include, for example, analytical reasoning, evaluating the reliability of knowledge, problem-solving, recognizing biases, considering various perspectives, and reaching well-reasoned conclusions (Shavelson et al. 2019). Additionally, the Delphi report,

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conducted by a panel of critical thinking experts, has been frequently used as a foundational reference for research of critical thinking (Facione 1990; see also Liu, Frankel, and Roohr 2014; Butler 2024). The report presents a consensus among experts on the theory of educational assessment and the instruction of critical thinking. According to the report, six core cognitive skills are essential: interpretation, analysis, evaluation, inference, explanation, and self-regulation (see Table 1). These skills are engaged in a holistic manner during critical thinking processes (Facione 1990; Hyytinen et al. 2021).

Additionally, the Delphi statement emphasizes that to make use of one's critical thinking, affective dispositions are also needed (see Table 1; Facione 1990). Affective dispositions refer to individuals' critical spirit and tendency to use cognitive skills for deciding what to believe and how to behave (Halpern 2014; Hyytinen et al. 2024). The critical spirit does not refer to a negative or hypercritical attitude towards everything and everyone. In turn, it means open-mindedness, flexibility in considering various alternatives, fair-mindedness, persistence, and honesty in facing one's own biases or egocentric tendencies (Facione 1990). Besides the definition of critical thinking, researchers have also debated whether this phenomenon is domain specific or domain general (Butler 2024; Shavelson 2018). Furthermore, although critical thinking is conceptualized in the literature as a multifaceted construct, it has been suggested that contemporary empirical research tends to focus more on specific skills rather than covering the complexity of this phenomenon (Bensley et al. 2016).

Earlier research on higher education has shown conflicting results regarding teaching and assessing critical thinking. On the one hand, systematic reviews have found that explicit instruction—such as opportunity for dialogue, the exposure of students to authentic or situated problems and examples, and mentoring—has positive effects on students' critical thinking skills (Abrami et al. 2015; El Soufi and See 2019). On the other hand, these same studies depict a research field that is

Table 1. Dimensions of critical thinking according to Facione (1990).

Dimension of critical thinking	Skill/disposition	Sub-skills/sub-dispositions
Cognitive skills	Interpretation	Categorization Decoding significance Clarifying meaning
	Analysis	Examining ideas Identifying arguments Analyzing arguments
	Evaluation	Assessing claims Assessing arguments
	Inference	Querying evidence Conjecturing alternatives Drawing conclusions
	Explanation	Stating results Justifying procedures Presenting arguments
	Self-regulation	Self-examination Self-correction
Dispositions	Approaches to life and living in general	E.g., inquisitiveness, being well-informed, alertness to opportunities to use critical thinking, self-confidence etc.
	Approaches to specific issues, questions or problems	E.g., clarity in stating the question, orderliness, diligence, reasonableness etc.

methodologically and theoretically fragmented (Abrami et al. 2015; El Soufi and See 2019). The internal consistency between the conception of critical thinking and its measurements has been found to be poor (Tiruneh, Verburgh, and Elen 2014). Additionally, a lack of replication leads to a situation in which research findings do not build up on each other (Tiruneh, Verburgh, and Elen 2014; El Soufi and See 2019). Consequently, the evidence for the instructional viability of promoting critical thinking remains limited. Therefore, it is not surprising that the research field has called for more robust measurements and research designs, as well as a more coherently aligned connection between theory and measurements (Abrami et al. 2015; El soufi and See 2019; Tiruneh, Verburgh, and Elen 2014). Additionally, it has been highlighted that that there is a need for measurements that capture the actual characteristics of critical thinking, not only students' perceptions and experiences related to learning critical thinking (Shavelson et al. 2019).

Performance-based assessments which aim to evoke authentic performance, covering elements of critical thinking through situations that resemble the real world, have been suggested as a solution to overcome the challenges of the assessments of critical thinking (Messick 1994; Shavelson et al. 2019). In completing the performance-based assessments tasks, students are asked to interpret, evaluate, and use evidence from multiple sources (Shavelson et al. 2019; Hyytinen et al. 2021). The common goal of the various performance-based assessments is to elicit what students know and can do (Messick 1994). Earlier research highlights that there are two main methods of performance-based assessments: open-ended performance tasks (PTs) and selected-response questions (Liu, Frankel, and Roohr 2014; Aloisi and Callaghan 2018). It has been suggested that the majority of performance-based assessments rely on selected-response questions to measure critical thinking (Liu, Frankel, and Roohr 2014; Zlatkin-Troitschanskaia, Shavelson, and Kuhn 2015). In open-ended performance tasks, students create their own answers to the assessment tasks, while in selected-response questions, students identify and select the correct answer from a list of given options (Aloisi and Callaghan 2018; Hyytinen et al. 2021; Liu et al. 2016). The weakness of selected-response questions is that excluding incorrect options or guessing the correct answer tends to be simpler than producing a response (Aloisi and Callaghan 2018; Butler 2024). A challenge of performance tasks is that they may be more vulnerable to biased scoring (Butler 2024). Both assessment methods typically involve materials that students analyze, but the response processes differ significantly (Hyytinen et al. 2021). Moreover, evidence suggests that these methods focus on different critical thinking skills (Messick 1994; Kleemola, Hyytinen, and Toom 2022). That is, performance tasks have been found to focus on several critical thinking skills and more holistic thinking processes while the selected-response questions, in turn, have been found to focus on individual skills (Shavelson et al. 2019; Zlatkin-Troitschanskaia et al. 2019; Kleemola, Hyytinen, and Toom 2022).

Additionally, Liu et al. (2016) have also argued that while selected-response questions have lower face validity, they have higher reliability compared to performance tasks for the same test time. However, validity is a multidimensional concept that encompasses not only face validity of the assessment or its internal structure (see Kleemola, Hyytinen, and Toom 2022) but also other essential aspects such as construct, content, and criterion validity (see Messick 1994). In performance-based assessments of critical thinking, particular attention should also be paid to cognitive validity, that is the extent to which the assessment tasks genuinely engage the intended cognitive processes (Hyytinen et al. 2021).

Although performance-based assessments, in general, seem to align better with the conceptualization of critical thinking than self-reports (Shavelson et al. 2019), little is known about how critical thinking has actually been conceptualized and how the used concepts are aligned with the measures in the current performance-based assessments of critical thinking in higher education. Yet, the conceptualization of critical thinking plays a crucial role in how it is assessed and the dimensions that are included in the assessment (Butler 2024; Zlatkin-Troitschanskaia, Shavelson, and Kuhn 2015). The present study seeks to fill this gap. The aim of this systematic-narrative literature review (Turnbull, Chugh, and Luck 2023) is to report how prior peer-reviewed empirical studies on performance-based assessments of critical thinking view and conceptualize critical thinking, what methods and frameworks are used, and what aspects of critical thinking are measured. In our view, these perspectives are crucial to understand and further develop critical thinking assessments.

The following research questions are addressed

1. What performance-based methods are used to assess critical thinking in the recently reported peer-reviewed articles?
2. How is the concept of critical thinking defined? What frameworks and theorizations are used?
3. What aspects of critical thinking are explored?

Materials and methods

Literature search

The review analysis consisted of four main phases: identification, screening, data extraction, and synthesis and analysis. The first three phases, representing the systematic component of the review, followed the PRISMA protocol, when applicable. The final phase was narrative in nature and involved qualitative content analysis. First, a literature search in the electronic databases of EBSCOhost (education and psychology databases), Scopus, and Social Science Database was conducted to identify peer-reviewed articles in English. The literature search was conducted in spring 2024. The publication year was limited to 2013–2023. We selected this ten-year period to ensure the review reflects the current state of research. By focusing on studies published between 2013 and 2023, we aimed to capture the most relevant and up-to-date research on performance-based assessments of critical thinking. The search was conducted in English. The searches were limited to the abstracts, keywords, and titles of the articles. Articles were selected for the review if they specifically focused on performance-based assessment of critical thinking in higher education. The main keywords utilized were 'critical thinking', 'higher education' and 'performance-based assessment' as well as their equivalents such as 'critical reasoning', 'university' and 'authentic assessment'. The literature search strategy is shown in Table 2. The searches resulted in 188 articles: 53 in EBSCOhost, 40 in Scopus and 95 in Social Science Database.

Table 2. Literature search strategy.

Databases	Keywords	Keywords (AND)	Keywords (AND)	Criteria for selecting articles
EBSCOhost Scopus Social Science	'critical thinking' OR 'critical reasoning' OR 'rational thinking' OR 'analytical thinking' OR 'rational reasoning' OR 'analytical reasoning'	'higher education' OR university OR college	'authentic assessment' OR 'performance assessment' OR 'performance-based assessment' OR 'performance task' OR 'constructed response task'	Focus on performance-based assessments of CT in higher education; written in English; Based on empirical evidence; Peer-reviews; Full texts; Published in 2013–2023

Screening

These articles were then imported to Rayyan, a web application used for screening articles in systematic reviews, and they were screened for duplicates. After removing duplicates ($n = 20$), the first and the second author read through the titles and abstracts to check the inclusion based on the criteria. The inclusion criteria were: (1) the article was published in a peer-reviewed journal, (2) the article reported empirical research of higher education students' critical thinking, distinguishing it from theoretical or conceptual studies, (3) the article employed performance-based assessment of critical thinking. Therefore, students' self-reports, review articles, policy articles, curriculum analysis, theoretical articles and the articles focusing on teachers' or employers' point of view were excluded. 51 articles were included after the first screening.

Data extraction

During the first round of data extraction, the first and second authors independently reviewed the articles to ensure they met the inclusion criteria. They each made individual decisions to include or exclude articles and then compared their decisions. Any discrepancies were discussed and resolved through negotiation. As a result, 20 articles were removed. In the second round of data extraction, all authors carefully read the remaining 31 articles. During this final phase, any articles that did not meet the inclusion criteria were removed. Ultimately, 28 articles were included in the analysis phase. The flow diagram of the articles included in the review is presented in Figure 1.

Synthesis and analysis

To address the research questions within this review, we conducted a qualitative content analysis (Elo et al. 2014). Each of the 28 studies was thoroughly analyzed. First, special attention was given to the methods and instruments used in assessing critical thinking. Subsequently, we synthesized the theoretical frameworks referenced in the theoretical background of the reviewed studies. Then, to develop a holistic understanding of the complex construct of critical thinking and to enable

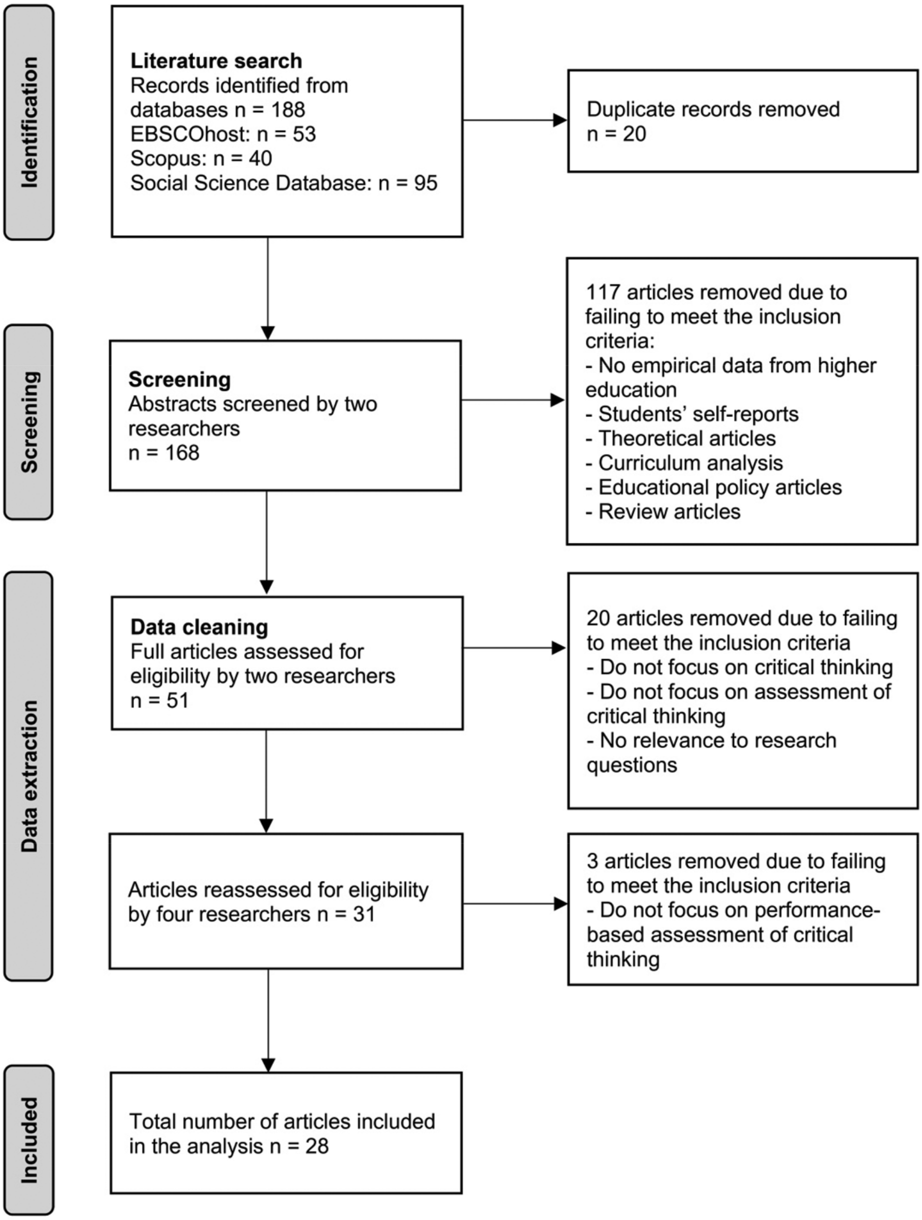

Figure 1. The flow diagram of articles included in review.

comparisons of research findings across different contexts, the next phases of the synthesis combined data-driven and theory-driven approaches. The analysis of the conceptions of critical thinking focused on three main aspects: skills, knowledge, and dispositions (cf. Facione 1990; Halpern 2014). Thereafter, we analyzed how performance-based assessments were operationalized by first identifying the assessed aspects in the reviewed articles. These aspects were then categorized using Facione's Facione (1990) framework, which includes interpretation, analysis, evaluation, inference, explanation, self-regulation, and dispositions. All four authors participated in the data analysis. To enhance the trustworthiness of the qualitative analysis, we used investigator triangulation (Denzin 2012).

Study characteristics

Of the 28 studies selected for the review, 24 utilized a cross-sectional research design. Six of the studies were pilot studies to test the effectiveness or validity of the assessment of critical thinking. Four of the studies employed an intervention, experimental or quasi-experimental design with control group (Hawthorne et al. 2015; Zabit, Karagiannidou, and Zachariah 2016; Cargas, Williams, and Rosenberg 2017; McGrew et al. 2019). The number of participants in the studies varied significantly, with the smallest with 5 and the largest with 120 915 participants. 43% of studies selected for the review were conducted in Europe, 32% in the USA, 14% in Asia, and 7% in Africa. One of the studies utilized a big data set from North and South America as well as Europe.

Results

Performance-based assessments of critical thinking – what assessment methods were used?

The 28 studies analyzed in this review employed a variety of methods to assess critical thinking. The most commonly reported performance-based assessment method was the open-ended performance task. Of the 28 articles reviewed, 18 utilized performance tasks or other written assignments. Six articles employed selected-response questions or Likert-type items (see Table 3). Additionally, four studies included a combination of performance task and selected-response tasks to capture critical thinking. The time limits for these assessments varied from approximately half an hour to fourteen days. Typically, performance tasks were allotted

Table 3. Assessment methods.

Methods	Studies
Open-ended performance task or written assignment tasks	CLA/ based on CLA: Hyytinen, Toom, and Postareff (2018), Hyytinen et al. (2024), Kleemola, Hyytinen, and Toom (2023), Mongkuo and Mongkuo (2017), Repo et al. (2017), Schendel & Tolmie (2016), Schendel (2015)
	IPAL: Hyytinen and Toom (2019), Shavelson et al. (2019), Zlatkin-Troitschanskaia et al. (2019), Zlatkin-Troitschanskaia et al. (2020)
	GEFSA: Rhodes et al. (2016), Rhodes, Danaher, and Kranov (2017), Rhodes, Danaher, and Kranov (2018)
	CAT: Cargas, Williams, and Rosenberg (2017)
	Written assignments: Hathcoat et al. (2016), McGrew et al. (2019), Sibulkin and Butler (2019)
Selected-response questions	ETS: Demeter, Robinson, and Frederick (2019), Hawthorne et al. (2015)
	CRT, Watson card selection: Mata et al. (2013)
	ERIT: Smith, Fulcher, and Sanchez (2017)
	CCTST: Zabit, Karagiannidou, and Zachariah (2016)
	Critical thinking in Psychology: Pawlow and Meinz (2017)
Combination of performance task and selected-response questions	CLA PT and MCQ: Kleemola, Hyytinen, and Toom (2022), Van Damme et al. (2023)
	CLA PT and GSA MCQ: Hyytinen et al. (2015), Nedelová and Šukolová (2017)

60-90 min, while selected-response tasks had stricter time limits, often around 30 min. In 11 studies, the time limit for the assessment was not reported.

The majority of the studies based their conclusions about critical thinking on students' performance task responses. 11 studies used performance tasks from the Council for Aid to Education's Collegiate Learning Assessments (CLA; Van Damme et al. 2023) or adaptations based on CLA. Four of the performance tasks were developed in the iPAL project (e.g.). Additionally, in three reviewed articles, the General Education Foundation Skills Assessment method and measurement tool (GEFSA; Rhodes et al. 2016; Rhodes, Danaher, and Kranov 2017; Rhodes, Danaher, and Kranov 2018) was used. One study used Critical Thinking Assessment Test (CAT; Cargas, Williams, and Rosenberg 2017). Additionally, the remaining three studies used researcher-developed written assignment tasks. A common feature of the performance tasks and assignments was their inclusion of real world, realistic or authentic problems to which students responded using source materials. Source materials varied from a few to 22 documents. In addition, four tasks had an unlimited online search (Rhodes et al. 2016; Rhodes, Danaher, and Kranov 2017; Rhodes, Danaher, and Kranov 2018; McGrew et al. 2019). Most of the performance tasks were individually completed assignments. The exception to this is the GEFSA, which was a group task of four to five students. In this task, students discuss online for 14 days about the problem.

In the reviewed studies (see Table 3), in total, ten studies administered selected-response questions in measuring critical thinking. Two studies used the CLA+ selected-response questions, two the Australian Council for Educational Research's Graduate Skills Assessment (GSA) test, and two relied on the Educational Testing Service's Tasks in Critical Thinking (ETS) to capture critical thinking. In turn, one study used the Ethical reasoning Identification Test (ERIT), which was a researcher-developed selected-response test. One study also administrated the California Critical Thinking Skills Test Form A (CCTST; Facione 1990) to measure critical thinking. In addition, one study used ten selected-response items from Critical thinking in Psychology by Lawson (1999) to measure critical thinking. There was also a study that used the Cognitive reflection task (CRT) and Watson card selections tasks to assess critical thinking.

How is critical thinking defined?

There was no consistent theory to which the reviewed studies referenced in their definitions. The most often referred researchers were Halpern ($n = 9$), Ennis ($n = 8$) and Facione (i.e. Delphi report; $n = 7$). Most of the studies ($n = 17$) defined critical thinking as a skill or a set of skills, while 11 studies emphasized the importance of disposition in addition to skills. There was large variation in definitions of critical thinking in those articles which determined critical thinking as a skill or a set of skills. Some of these studies provided detailed definitions, including analyzing, evaluating and applying information to make a decision or solve a problem (Smith, Fulcher, and Sanchez 2017; Hyytinen, Toom, and Postareff 2018; Demeter, Robinson, and Frederick 2019; McGrew et al. 2019; Shavelson et al. 2019; Sibulkin and Butler 2019; Zlatkin-Troitschanskaia et al. 2020). In certain studies, critical thinking was determined as one of the generic skills or transferable skills without providing a more detailed definition of critical thinking or what critical thinking includes (Rhodes et al. 2016; Rhodes, Danaher, and Kranov 2017; Rhodes, Danaher, and Kranov 2018; Van Damme et al. 2023). Additionally, three studies did not define critical thinking at all (Mata et al. 2013; Hathcoat et al. 2016; Pawlow and Meinz 2017). The knowledge aspect of critical thinking was difficult to identify because it is closely intertwined with skill, and in most studies, it was not specifically mentioned or defined. In the 11 studies where dispositions were mentioned, the definitions of critical thinking were generally more detailed and comprehensive, even though disposition itself was not specifically defined in these studies. These studies emphasized that critical thinking is a set of cognitive skills and affective dispositions or self-regulatory skills (Hyytinen et al. 2015; Schendel 2015; Schendel and Tolmie, 2016; Cargas, Williams, and Rosenberg 2017; Nedelová and Šukolová 2017; Repo et al. 2017; Hyytinen and Toom 2019; Zlatkin-Troitschanskaia et al. 2019; Kleemola, Hyytinen, and Toom 2022; Hyytinen et al. 2024; Kleemola, Hyytinen, and Toom 2023).

Aspects of critical thinking measured

Almost all of the studies defined which dimensions or skills of critical thinking were measured. Two articles stated only that they measured critical thinking (Hawthorne et al. 2015, Pawlow and Meinz 2017); thus, they are not included in the following analysis. The descriptions of the measured skills varied greatly from one-word mentions to entire scoring rubrics. Thus, investigative interpretation has been necessary. Where available, wider definitions have been used in reflecting on the measured skill's position in Facione's framework.

All six skills listed as core critical thinking skills by Facione (1990; see also Table 1) were investigated in some of the reviewed articles. However, none of the articles focused on all six skills, and none of the skills were present in all articles. One article did not focus on any of the core critical thinking skills. The measurement of the skills in each article is presented in Table 4. None of the articles focused on dispositions in their performance-based assessment.

Interpretation was measured in eleven articles. Examples of the measured interpretative skills are organizing information (Hyytinen and Toom 2019), identification of data sources (Zlatkin-Troitschanskaia et al. 2020), identification of the problems (Hathcoat et al. 2016), and recognition of evidence (Zlatkin-Troitschanskaia et al. 2020).

Analysis was measured in seven articles. Examples of the measured analytic skills are evaluating arguments (Shavelson et al. 2019), synthesizing results/information (Cargas, Williams, and Rosenberg 2017; Hyytinen and Toom 2019; Zlatkin-Troitschanskaia et al. 2020), and ability to determine if information is relevant (Schendel 2015; Schendel and Tolmie 2016).

Evaluation was measured in 21 articles. Examples of the measured evaluative skills are evaluating and using relevant information (Rhodes, Danaher, and Kranov 2018; Rhodes, Danaher, and Kranov 2017), avoiding judgmental errors (Hyytinen and Toom 2019), assessing quality and relevance of evidence (Mongkuo and Mongkuo

Table 4. Dimensions and skills that are measured in the reviewed articles.

	Cognitive skills							
	Interpretation	Analysis	Evaluation	Inference	Explanation	Self-regulation	Dispositions	Other
Cargas, Williams, and Rosenberg (2017)	x	x	x		x	x		
Demeter, Robinson, and Frederick (2019)	x		x	x				
Hathcoat et al. (2016)	x		x		x			
Hyytinen et al. (2015)				x	x			
Hyytinen, Toom, and Postareff (2018)				x	x			
Hyytinen et al. (2024)				x	x			WM[1]
Hyytinen and Toom (2019)	x	x	x	x	x			E[2]
Kleemola, Hyytinen, and Toom (2022)			x	x	x			WM [1]
Kleemola, Hyytinen, and Toom (2023)				x	x			WM [1]
Mata et al. (2013)			x			x		
McGrew et al. (2019)			x					
Mongkuo and Mongkuo (2017)		x	x					
Zabit, Karagiannidou, and Zachariah (2016)			x	x				
Nedelová and Šukolová (2017)			x	x	x			
Repo et al. (2017)			x	x	x			
Rhodes et al. (2016)	x		x	x				G[3], L[4]
Rhodes, Danaher, and Kranov (2017)	x		x	x				G[3], L[4]
Rhodes, Danaher, and Kranov (2018)	x		x	x				G[3], L[4]
Schendel (2015)		x	x	x				S[5]
Schendel and Tolmie (2016)	x	x	x					S[5]
Shavelson et al. (2019)	x	x	x		x			
Sibulkin and Butler (2019)			x					
Smith, Fulcher, and Sanchez (2017)								E[2]
Van Damme et al. (2023)			x	x	x			WM [1]
Zlatkin-Troitschanskaia et al. (2019)	x		x	x	x			WM [1]
Zlatkin-Troitschanskaia et al. (2020)	x	x	x	x	x			
Total	11	7	21	17	14	2	0	12

[1]WM: writing mechanics.
[2]E: ethics.
[3]G: group skills.
[4]L: language skills.
[5]S: statistical skills.

2017), ability to evaluate if data supports an argument (Schendel 2015; Schendel and Tolmie 2016), and evaluating evidence (McGrew et al. 2019).

Inference was measured in 17 articles. Examples of the measured inference skills are analytical reasoning (Nedelová and Šukolová 2017), reasoning based on evidence (Zlatkin-Troitschanskaia et al. 2020), scientific and quantitative reasoning (Kleemola, Hyytinen, and Toom 2022), logical and deductive reasoning (Demeter, Robinson, and Frederick 2019), and inductive and deductive inference (Zabit, Karagiannidou, and Zachariah 2016).

Explanation was measured in 14 articles. Examples of the measured explanative skills are writing effectiveness (Hyytinen et al. 2015; Nedelová and Šukolová 2017; Kleemola, Hyytinen, and Toom 2022; Hyytinen et al. 2024; Kleemola, Hyytinen, and Toom 2023), integrating and structuring arguments (Shavelson et al. 2019), writing arguments and counterarguments (Hyytinen and Toom 2019), presentation of student's own position (Hathcoat et al. 2016), and make claims (Cargas, Williams, and Rosenberg 2017).

Self-regulation was measured in two articles. In these studies, students were required to critically evaluate their own performance (Cargas, Williams, and Rosenberg 2017) and assess their own reasoning in relation to that of others (Mata et al. 2013).

None of the reviewed articles explicitly expressed dispositions in their targets of assessment. In twelve articles, other skills than the ones listed in Facione's (1990) framework were measured. These skills were writing mechanics (5 articles), group skills (3 articles), language skills (3 articles), statistical skills (2 articles), and ethical skills (2 articles).

Discussion and conclusion

This study set out to investigate the performance-based methods used to assess critical thinking in recently reported peer-reviewed articles, identify the frameworks and concepts employed, and explore the various aspects of critical thinking examined. Across the 28 studies, we found that the most common assessment method in this research field was open-ended performance tasks. Interestingly, one-third of the studies relied solely on selected-response questions or combined selected-response with performance tasks. The results suggest that the CLA or adaptations based on CLA performance tasks dominate current research. However, there was more variety in the tasks used for selected-response assessments. Overall, the limited number of published studies, coupled with the high number of CLA-based studies, indicates that the research field of performance-based assessment of critical thinking is rather narrow. While previous literature has suggested that most assessments of critical thinking rely on selected-response formats (Liu, Frankel, and Roohr 2014; Zlatkin-Troitschanskaia, Shavelson, and Kuhn 2015), our review of peer-reviewed empirical studies reveals a stronger emphasis on performance tasks, indicating a shift in focus within published research. Additionally, the issue of research design must be mentioned. The majority of the studies included in this review employed a cross-sectional design. Only a few utilized quasi-experimental designs with pretests and posttests. Given the central role of critical thinking in higher education and the limited use of performance-based methods, this study highlights a clear need for more research grounded in such approaches within this field.

This review revealed that most studies outlined an explicit definition of critical thinking. In most cases, critical thinking was described as a skill or a set of skills, whereas a smaller number of studies emphasized the importance of disposition or knowledge in addition to skills. A few of the studies did not define critical thinking at all in their theory section. Thus, the results indicated that there was no consistent theoretical framework to which the reviewed studies referenced in their definitions. The most often referenced theoretical framework was the Delphi report (Facione 1990). Again, this involved only a limited number of studies. These results highlight that skills (i.e. cognitive processes) are emphasized in the current definitions of performance-based assessment studies, without taking into account the multidimensionality and complexity of the phenomenon (cf. Bensley et al. 2016; Butler 2024; Hyytinen et al. 2024).

We found that almost all of the studies specified which skills of critical thinking were measured. However, the descriptions varied extensively: from one-word mentions to entire scoring rubrics. Additionally, in two studies, both based on selected-response tasks, the measured skills were not explicitly mentioned at all. All six core critical thinking skills identified by Facione (1990) were investigated in some of the reviewed articles. However, none of the studies examined all six skills comprehensively, and no single skill was present in every article. Additionally, self-regulation was only addressed in two studies (cf. Bensley et al. 2016; Hyytinen et al. 2024). Furthermore, none of the studies included dispositions in their performance-based assessments. The findings indicate that evaluation and inference were the most frequently measured skills in the reviewed studies. It was not surprising that evaluation was the most often measured skill in the reviewed articles. Critical evaluation of evidence can be characterized as central to critical thinking, as defined by the narrowest interpretations of the concept.

Overall, the results suggest that the studies analyzed in this review predominantly focused on a limited number of skills. We detected no differences between performance tasks and selected-response tasks in the number of skills they measured. However, our analysis did not reveal whether the skill combinations were measured holistically (as performance tasks tend to do) or targeting skills individually (as selected-response tasks tend to do). In the literature, critical thinking is seen as a multifaceted construct (Facione 1990; Halpern 2014), but current empirical studies tend to emphasize specific skills more than a comprehensive understanding of this phenomenon (Bensley et al. 2016). From a validity perspective, our findings raise concerns particularly regarding construct and cognitive validity, as the current assessments may not fully capture the multifaceted nature of critical thinking. Therefore, in the future, multi-method research is needed to better capture the multifaceted complexities within critical thinking (Liu, Frankel, and Roohr 2014; Zlatkin-Troitschanskaia, Shavelson, and Kuhn 2015). Furthermore, although the validity of the articles was not the primary focus of this study, we observed during the course of our analysis that validity was seldom addressed in the reviewed studies. This observation highlights an important gap, and we suggest that future research should give more attention to this aspect (see also Aloisi and Callaghan 2018; Kleemola, Hyytinen, and Toom 2022; Liu, Frankel, and Roohr 2014).

This study is not without limitations. One limitation that could have influenced the findings is that our analysis of the measured aspects of critical thinking required interpretation. The concepts used in the reviewed articles were not always clearly defined. Additionally, it was notable that the reviewed studies used terms such as 'analysis' in various ways that did not necessarily align with Facione's (1990) definition. This issue is particularly relevant to the third research question. Another limitation is that the inclusion criteria excluded self-report studies. Dispositions may be easier to capture using self-report instruments, which could explain why dispositions were not prominently addressed in the articles included in our analysis. Furthermore, the findings of this systematic review are constrained by the selected studies that were available at the time of review. As new studies are published, they may provide additional insights or alter current findings and conclusions.

Based on the findings, several theoretical and practical implications for researchers can be outlined. Future research should employ a broader range of performance-based assessments of critical thinking. Incorporating a variety of tasks and other innovative performance-based assessment methods could provide a more comprehensive evaluation of critical thinking. Additionally, future studies should also include the performance-based assessment of dispositions and knowledge alongside skills. Understanding the role of dispositions and knowledge in critical thinking provides a more holistic view of how these skills are developed and applied. Furthermore, researchers should aim to clearly define the concepts and terms used in their studies (cf. Tiruneh, Verburgh, and Elen 2014), ensuring consistency with established definitions, for example those by Facione (1990). Such challenges in comparison and accumulation of knowledge seem to be quite common in current research on critical thinking (Tiruneh, Verburgh, and Elen 2014; El Soufi and See 2019) and in general in the educational research (e.g. Dias-Broens, Meeuwisse, and Severiens 2024). Moreover, there is a need for novel theoretical development of critical thinking in the research field. Since the Delphi report (Facione 1990), there has been little theorizing in the field. Clear and established theorization will help in comparing findings and accumulating knowledge across different studies. This will also help to advance the understanding and application of critical thinking in contemporary higher educational contexts. The findings of this study suggest that there is also a need for more longitudinal and quasi-experimental studies with pretests and posttests. These designs can provide deeper insights into the development of critical thinking skills over time and the effectiveness of different instructional methods. In addition, artificial intelligence (AI) could be used to provide timely and personalized formative feedback, which would support students' learning processes more effectively. Furthermore, AI could be employed to support the scoring of open-ended performance tasks, which are often resource-intensive due to the need for human scorers (Liu, Frankel, and Roohr 2014). By addressing these practical implications, future research can contribute to a more robust and cohesive understanding about critical thinking in higher education.

Conclusions

This study highlights that in performance-based assessments of critical thinking, current empirical research focuses on open-ended performance tasks, with a significant reliance on CLA-based methods. The research field appears narrow, with a lack of consistent theoretical frameworks across the studies. Most studies conceptualized critical thinking as a set of skills, neglecting the role of dispositions. In a similar vein, the knowledge aspect was not explicitly addressed. The skill-based perspective was also emphasized in the measured aspects, with evaluation and inference as the most frequently measured skills. The outcomes revealed a mismatch between the definition of critical thinking and the ways it is defined and measured in performance-based assessments. The findings, thus, highlight the need for more holistic performance-based assessment methods, including the assessment of self-regulation, dispositions and knowledge to better capture the multifaceted nature of critical thinking (Hyytinen et al. 2024; Liu et al. 2016; Zlatkin-Troitschanskaia, Shavelson, and Kuhn 2015). Future research should also aim for clearer definitions and theorization of critical thinking and more longitudinal and quasi-experimental designs to better understand the development of critical thinking over time.

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